Internship Report – Day 12

Date: (Mention Date)
Department: Cloud & Networking
Intern Name: Syed Fuzail Abdullah

Topic Covered: Networking Fundamentals, CIDR Calculation, Subnetting, and VNet Design

On Day 12 of my internship, I learned fundamental networking concepts including IP addressing, CIDR notation, subnetting, host bit calculation, and decimal-to-binary conversion. These concepts are essential for designing secure and scalable cloud networks in platforms such as AWS and Microsoft Azure.

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1. Introduction to Networking

Networking enables communication between devices using unique identifiers known as **IP addresses**. Each IP address consists of:

- **Network Portion** – Identifies the network
- **Host Portion** – Identifies devices within the network

Example:

192.168.1.10/24

- `/24` represents CIDR notation
- First 24 bits → Network
- Remaining 8 bits → Host

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2. IP Address Classes

IP addresses are categorized into different classes:

Class	Range	Default Subnet Mask	Usage

A	0 – 126	255.0.0.0	Large networks
B	128 – 191	255.255.0.0	Medium networks
C	192 – 223	255.255.255.0	Small networks
D	224 – 239	N/A	Multicast
E	240 – 255	N/A	Experimental

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3. CIDR (Classless Inter-Domain Routing)

CIDR notation defines how many bits belong to the network.

Example:

192.168.1.0/26

Formulas Learned

- **Host Bits = 32 − CIDR**
- **Total IP Addresses = 2^n**
- **Usable Hosts = $2^n - 2$**

(where n = number of host bits)

Example:

/24 → Host bits = 8
2^8 = 256 total IPs
256 − 2 = 254 usable hosts

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4. Subnetting

Subnetting divides a large network into smaller logical networks for better management and security.

Example:

192.168.1.0/24 divided into /26

Creates 4 subnets:

- 192.168.1.0/26
- 192.168.1.64/26
- 192.168.1.128/26
- 192.168.1.192/26

Each subnet:

2^6 = 64 total IPs
64 − 2 = 62 usable hosts

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5. Decimal to Binary Conversion

Each IP address contains four octets (8 bits each).

Example:

192.168.1.1

Binary form:

Decimal	Binary
192	11000000
168	1010100 0
1	0000000 1
1	0000000 1

Final Binary Representation:

11000000.10101000.00000001.00000001

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6. Practical Task – Virtual Network (VNet) Design

As part of the practical exercise, we were assigned a task to design and plan a **Virtual Network (VNet)** with an optimal IP address range and create multiple subnets based on required host capacity.

Requirement Given

Create a VNet containing three subnets:

Subnet Name	Required IPs
Test	30 IP addresses
Dev	30 IP addresses
Cloud	60 IP addresses

Subnet Calculation

To determine subnet sizes, CIDR calculations were performed:

✅ **Subnet for 30 Hosts**

Formula:

$2^n - 2 \geq 30$

$2^5 = 32$ addresses
$32 - 2 = 30$ usable hosts

Required subnet:

/27 (32 IP addresses)

So:

- **Test subnet → /27**
- **Dev subnet → /27**

✅ **Subnet for 60 Hosts**
$2^6 = 64$ addresses

64 − 2 = 62 usable hosts

Required subnet:

/26 (64 IP addresses)

So:

- **Cloud subnet → /26**

Example VNet Address Plan

Chosen VNet range:

10.0.0.0/24

Subnet allocation:

Subnet	CIDR	IP Range
Cloud	10.0.0.0/26	10.0.0.0 – 10.0.0.63
Test	10.0.0.64/27	10.0.0.64 – 10.0.0.95
Dev	10.0.0.96/27	10.0.0.96 – 10.0.0.127

This planning ensured:

- Efficient IP utilization
- Proper network segmentation
- Future scalability

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7. Key Concepts Learned

- CIDR calculation and prefix understanding
- Host bits and usable host calculation
- Subnetting techniques
- Binary conversion of IP addresses
- Designing VNets using real-world requirements

- IP planning for cloud environments

Conclusion

Day 12 of the internship strengthened my understanding of networking fundamentals and practical subnet design. I learned how to calculate CIDR ranges, determine host requirements, and design a Virtual Network with multiple subnets based on real-world scenarios. These concepts are essential for cloud architecture, secure infrastructure deployment, and network segmentation in AWS and Azure environments.

Here is your **updated Day 12 Internship Report** with the additional practical task included professionally:

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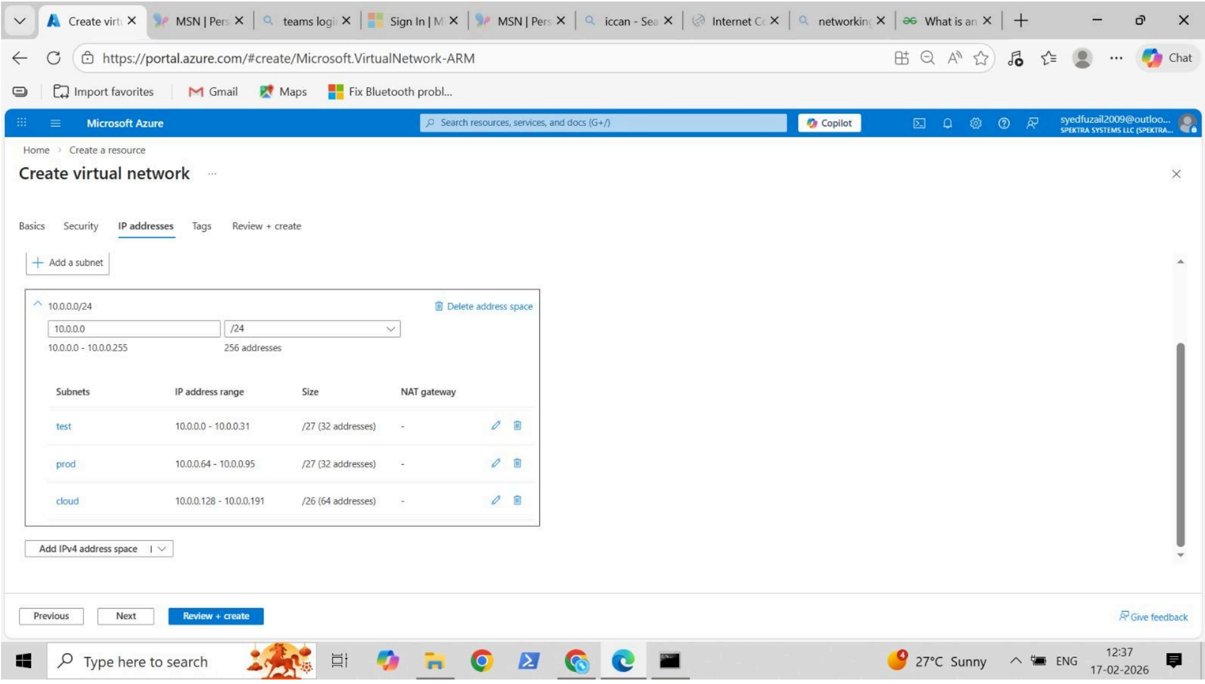
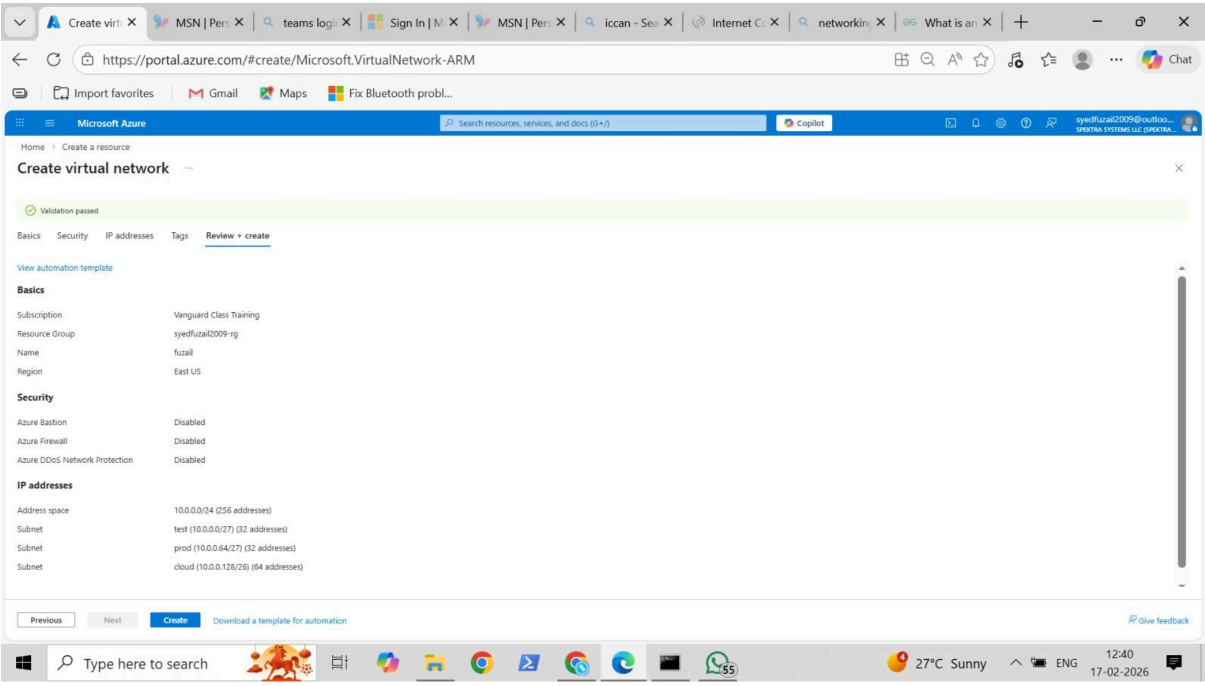
If you want, I can next make a **more architect-level version** (which impresses internship evaluators more) or format it exactly like your previous Day 9 report style.


https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM
Microsoft Azure
Home > Create a resource
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Add a subnet
10.0.0.0/24
Delete address space
10.0.0.0
/24
10.0.0.0 - 10.0.0.255
256 addresses
Subnets
IP address range
Size
NAT gateway
test
10.0.0.0 - 10.0.0.31
/27 (32 addresses)
-
prod
10.0.0.64 - 10.0.0.95
/27 (32 addresses)
-
cloud
10.0.0.128 - 10.0.0.191
/26 (64 addresses)
-
Add IPv4 address space
Previous
Next
Review + create
Give feedback
Type here to search
27°C Sunny
ENG
12:37
17-02-2026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Microsoft Azure
Home > Create a resource
Create virtual network
Validation passed
Basics
Security
IP addresses
Tags
Review + create
View automation template
Basics
Subscription
Vanguard Class Training
Resource Group
syedfuzail2009-rg
Name
fuzail
Region
East US
Security
Azure Bastion
Disabled
Azure Firewall
Disabled
Azure DDoS Network Protection
Disabled
IP addresses
Address space
10.0.0.0/24 (256 addresses)
Subnet
test (10.0.0.0/27) (32 addresses)
Subnet
prod (10.0.0.64/27) (32 addresses)
Subnet
cloud (10.0.0.128/26) (64 addresses)
Previous
Next
Create
Download a template for automation
Give feedback
Type here to search
27°C Sunny
ENG
12:40
17-02-2026